Prajwal Pisal

Computational Materials Researcher

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Professional Summary

Computational materials scientist and doctoral researcher with 4+ years of experience at the intersection of density functional theory (DFT), machine learning, and high-throughput materials discovery. Skilled in building end-to-end computational workflows, automating simulations on HPC infrastructure, and collaborating with experimental teams to translate model predictions into validated materials. Experienced in AI for materials science, quantum chemistry, and materials databases, with publications in high-impact journals such as *npj Computational Materials*. Seeking a role where I can design robust DFT-based workflows, integrate AI/ML for accelerated screening and decision-making, and contribute to closed-loop materials discovery.

Technical Skills

Computational Modeling	Density Functional Theory (DFT), Molecular Dynamics (MD), Crystalline Materials, Surface Catalysis, Thermodynamics and Phase Stability, High-Throughput Screening, Machine-Learned Force Fields (MLFFs)
DFT Tools	VASP, FHI-aims, Quantum Espresso, Gaussian09, Materials Project Database, Atomate, ASE, Pymatgen, Custodian
Machine Learning	Regression, Atomistic Descriptors, Deep Learning, Graph Neural Networks (GNNs), Model Finetuning and Evaluation, Feature Engineering, Data Processing and Visualization
Programming	Python, PyTorch, SciPy, Scikit-Learn, Pandas, XGBoost, Git, LaTeX
Computing and Workflows	High-Performance Computing (HPC), Slurm Scheduling, Docker and Containerization, Linux, Bash, MongoDB

Professional Experience

Doctoral Researcher — *Jan 2025 – Present*
Technical University of Munich (TUM) — *Munich, Germany*

- Continued doctoral research at TUM in close collaboration with experimentalists at Aalto University and strengthened computational-experimental partnerships across leading European universities.
- Built high-throughput workflows that use machine-learned force fields (e.g., models from the Fairchem and PET-MAD) to screen materials from the Materials Project database for CO_2 catalysis.
- Generated targeted DFT datasets and fine-tuned MLFFs to improve inference across oxide materials for more accurate evaluation of candidates for catalysis.

Visiting Researcher — *May 2023*
Carnegie Mellon University (CMU) — *Pittsburgh, United States*

- Collaborated with researchers at CMU and Meta Research to adapt and evaluate Fairchem GNNs for CO_2 catalysis, benchmarking their performance on target materials and integrating them into closed-loop workflows for materials discovery.

Doctoral Researcher — *Aug 2022 – Dec 2024*
Aalto University — *Espoo, Finland*

- Designed an ML-accelerated workflow for catalyst descriptor design and screening, leading to the identification of novel catalyst candidates published in *npj Computational Materials*.
- Leveraged high-performance computing resources at the Finnish national computing center to run large-scale DFT simulations (e.g., VASP, FHI-aims) and to train materials-specific ML models.
- Collaborated with synthesis and characterization teams to build and curate structured catalyst databases hosted on Zenodo and published in *Nature Scientific Data*, enabling robust ML training and reproducible experimentation.

Research Intern — *Jun 2021 – May 2022*
CSIR - National Chemical Laboratory — *Pune, India*

- Designed and executed DFT workflows on nanocluster catalysts for CO_2 conversion to analyze structure-activity relationships and identify promising active sites.
- Modeled and computed 2D materials for CO_2 catalysis using DFT (Quantum ESPRESSO).

Education

PhD in Machine Learning for Materials Discovery — *Technical University of Munich*
Aug 2022 – Dec 2026 (expected)
Started at Aalto University and continued at TUM following the relocation of the research group.
Thesis (working title): AI-Guided Materials Design for CO_2 Catalysis.

Integrated B.Sc.-M.Sc. in Natural Sciences — *Indian Institute of Science Education and Research Tirupati*
2017 – 2022
Specialization in Chemistry with interdisciplinary background in Physics, Mathematics, and Data Science.
Master's thesis: Discerning the factors of CO_2 activation on isolated and supported subnanometer copper clusters. GPA: 8.4/10.

Selected Conference Presentations

Research talk, **20^{th} International Conference for CO_2 Utilization 2023 (Bari, Italy)**: *"ML-driven efficient experimental screening for CO_2 to methanol conversion catalysts."*

Research talk, **Annual Conference of the Deutsche Physikalische Gesellschaft 2024** (Berlin, Germany): "ML-accelerated descriptor design for CO_2 catalysis."

Contributed research talk, **Psi-K Conference 2025** (Lausanne, Switzerland): "Descriptor design for CO_2 to methanol conversion based on work published in *npj Computational Materials."*

Selected Publications & Research Impact

Pisal P., Krejčí O., Rinke P. (2025). Machine learning accelerated catalyst descriptor design for catalyst discovery in CO_2 to methanol conversion. *npj Computational Materials*

Toldy A., **Pisal P.**, Krejčí O., Rinke P., Santasalo-Aarnio A (2025). TheMeCat: Dataset of Thermocatalytic Conversion of CO_2 to Methanol. *Nature Scientific Data*

Languages

English	Professional	**German**	B2
Hindi	Professional	**Marathi**	Native
Finnish	Basic		

Awards

INSPIRE Scholarship – 2017	Government of India – Top 1% students
Physics Days Conference, Finland – 2023	Poster honourable mention

Referees

Prof. Patrick Rinke	Professor, Technical University of Munich (TUM) (*patrick.rinke@tum.de*)
Dr. Ondřej Krejčí	Postdoctoral Research Fellow, Aalto University (*ondrej.krejci@aalto.fi*)